

RESEARCH ARTICLE

Genetic algorithm-based structural optimization of foldable OLED panels

Qibin Feng  | Chen Zheng, SID Student Member |
Zhe Dong, SID Student Member | Yusong Guo, SID Student Member | Zi Wang |
Guoqiang Lv 

Special Display and Imaging Technology
Innovation Center of Anhui Province,
National Engineering Laboratory of
Special Display Technology, Hefei
University of Technology, Hefei, China

Correspondence

Guoqiang Lv, Special Display and
Imaging Technology Innovation Center of
Anhui Province, National Engineering
Laboratory of Special Display Technology,
Hefei University of Technology, Hefei
230009, China.
Email: guoqianglv@hfut.edu.cn

Funding information

Major Science and Technology Research
Project of Anhui Province, Grant/Award
Number: 2023z020002

Abstract

OLEDs are playing an important role in flexible displays, smart wearable, in-vehicle displays, and other fields. Foldable OLED panels consist of multiple layers of film. To ensure the reliability, some key films such as the OLED should be placed on the neutral plane, which is usually achieved by adjusting the thickness of each layer. In this paper, genetic algorithm is introduced to find the optimum configurations of the thickness of each layer. Based on the mathematical express of the stress happening on each film when an OLED panel is folded, the objective function can be defined and the optical thicknesses of some layers can be got. The simulation results show that with such optical thickness configuration, the stresses on key films can effectively reduce. The method proposed in the paper can help improve the folding performance of a foldable OLED panel.

KEYWORDS

foldable OLED, genetic algorithm, neutral plane, stress, structural optimization

1 | INTRODUCTION

In recent years, flexible organic light-emitting devices (OLEDs) have become more and more popular.^{1,2} An OLED panel is a composite stacked structure composed of different thin functional films.^{3,4} During the process of folding, excessive stresses on some key films can lead to the functional failure, which can affect the overall reliability of the foldable panel.⁵ According to the neutral plane theory, if the key films can be placed on the neutral plane, the stress on this film can be greatly reduced, which can improve the overall structural reliability.⁶

Many scholars have shown interest in the neutral plane theory and have applied it to the research on foldable OLED panels. Park et al. added a layer of aluminum buffer between the ITO film and the substrate, which makes the neutral plane happen on ITO film to reduce

the stress on ITO film.⁷ Lee et al. changed the cover thickness of the foldable OLED device to make the neutral plane move towards the ITO film, thus reducing the stress on the ITO film.⁸ Niu et al. adjusted the thickness of the encapsulation layer in AMOLED panels to reduce the stress on the barrier layer locating above the OLED layer, while keeping the neutral plane still on the OLED layer.⁹ By adjusting the ratio of the thicknesses of the top and bottom layers of the films, Li's research team demonstrated that the neutral plane can happen on the layer with the larger thickness.⁵

The above researches indicate that the reliability can be improved by moving neutral plane towards the key film by adjusting the thickness of each layer in foldable OLED panels. Currently, finite element analysis is usually used to analyze the stress distribution of a foldable OLED panel. Based on huge simulations, the optimal

configuration can be possibly found in which the neural plane happens on the key film.^{10,11} The current simulation based on analysis method suffers from the problems of large workload, long optimization time, and difficulty in finding the optimal solution. In this paper, genetic algorithm is introduced to find the optimum configurations of the thickness of each layer, which can help shorten the design time and reduce development risk.

2 | MODEL OF FOLDABLE OLED PANELS

The stacked structure of a foldable OLED panel is illustrated in Figure 1, which consists of cover layer, glue layer, passivation layer, cathode, OLED layer, ITO layer, three barrier layers, and PI substrate.

As the fact that the panel is symmetrical in structure, only half of the panel is analyzed. The schematic diagram of the structure before and after being folded is shown in Figure 2, which is inward folding with the rotation angle of 90 degrees. To realize the folding action, the rigid body is necessary, which is also included in Figure 2. The length of the foldable OLED panel and the length of the rigid body are denoted as L , L_R , and R is the bending radius, respectively.

3 | GENETIC-ALGORITHM-BASED STRUCTURAL OPTIMIZATION

As the fact that there exists almost infinite number of thickness combinations, it is almost impossible to simulate all the combinations to find an optimal solution. Take the combination of three layers (Cover, Glue, and PI) as example. Even there is only 20 thickness choice for each layer, the total simulation number would reach 8000, which is almost impossible to finish and make

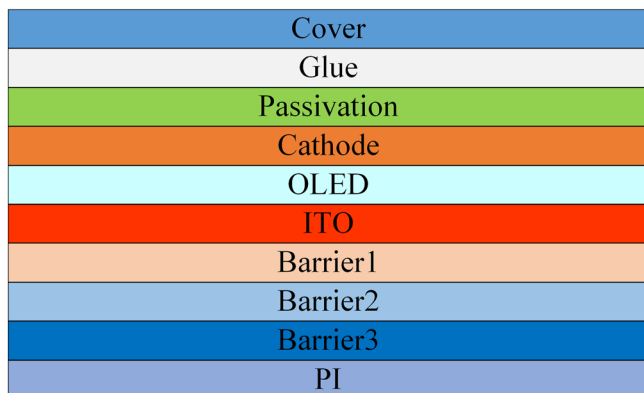


FIGURE 1 Stacked structure of a foldable OLED panel.

genetic algorithm become one inspiration to be used in this paper.

Genetic algorithm is an optimization algorithm that simulates the principles of biological genetic evolution. It combines the principles of “survival of the fittest” and genetic crossover and mutation, encoding the problem to be solved into chromosomes. By simulating the process of biological evolution and performing population iteration, selection, crossover, mutation, and multiple iterations, the algorithm selects the best chromosomes. The solution corresponding to the best chromosome is the optimal solution to the problem.

When employing the genetic algorithm to optimize the thickness of each layer in foldable OLED panels to make the neutral planes happening on some key films, the reasonable objective function should be firstly established.

Lee's research team has provided a mathematical calculation model for the bending stress experienced by each film in the multilayer stacked structure during bending.⁵ The equations for the stress experienced on each film when bending is expressed as follows:

$$d_1 + d_2 + \dots + d_{n-1} \leq z_n \leq h: \sigma_n = E_n \left[\epsilon_{bot} + \frac{z}{h} (\epsilon_{top} - \epsilon_{bot}) \right] \quad (1a)$$

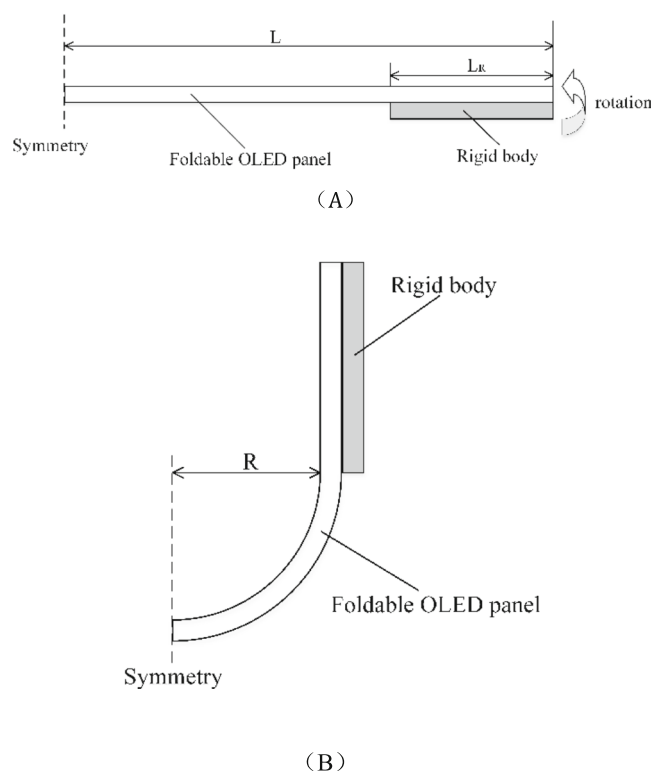


FIGURE 2 The schematic diagram of the foldable OLED panel: (A) before being folded, (B) after being folded.

$$d_1 \leq z_2 \leq d_1 + d_2 : \sigma_2 = \bar{E}_2 \left[\varepsilon_{bot} + \frac{z}{h} (\varepsilon_{top} - \varepsilon_{bot}) \right] \quad (1b)$$

$$0 \leq z_1 \leq d_1 : \sigma_1 = \bar{E}_1 \left[\varepsilon_{bot} + \frac{z}{h} (\varepsilon_{top} - \varepsilon_{bot}) \right] \quad (1c)$$

$$\varepsilon_{bot} = -\frac{1}{2R} \left[\frac{\sum_{i=1}^n \bar{E}_i (z_i^2 - z_{i-1}^2)}{\sum_{i=1}^n \bar{E}_i (z_i - z_{i-1})} \right] \quad (2)$$

$$\varepsilon_{top} = \frac{h}{R} - \frac{1}{2R} \left[\frac{\sum_{i=1}^n \bar{E}_i (z_i^2 - z_{i-1}^2)}{\sum_{i=1}^n \bar{E}_i (z_i - z_{i-1})} \right] \quad (3)$$

$$Z_i = \sum_{j=1}^i d_j \quad (4)$$

$$\bar{E}_i = \frac{E_i}{1 - \nu_i^2} \quad (5)$$

where E_i represents elastic modulus of each film, ν_i represents Poisson's ratio of each film, h is the thickness of the overall foldable panel, z_i represents the position of each film in the thickness direction, R is the bending radius, and ε_{top} and ε_{bot} represent the bending strain at the top and bottom surfaces of the panel.

Based on the above calculation equations, we can calculate the bending stresses on each film during the bending process. First, with the consideration of no changing on the coating process of OLED, the thicknesses of the OLED layer, barrier layers, and passivation layer are not to be optimized. Therefore, the thicknesses of the cover layer (d_{Cover}), glue layer (d_{Glue}), and PI substrate (d_{PI}) are selected as the optimized parameters. After that, the objective function is established.

In a foldable OLED panel, the luminance and uniformity are mainly decided by OLED layer, and the layer of OLED is taken as a key functional layer. The reduction of the stress on OLED layer is beneficial for enhancing the overall reliability of the panel. Therefore, we choose the OLED layer as the stress improvement objective, and thus the objective function F is defined as following:

$$F = \bar{E}_{OLED} \left[\varepsilon_{bot} + \frac{z_{OLED}}{h} (\varepsilon_{top} - \varepsilon_{bot}) \right] \quad (6)$$

$$\bar{E}_{OLED} = \frac{E_{OLED}}{1 - \nu_{OLED}^2} \quad (7)$$

Equation (6) represents the stress value on OLED layer, as smaller as better. Based on this objective function, the optimal combination of the thickness of the cover layer, glue layer, and PI substrate can be achieved.

When we need to improve the stress distribution of multiple layers, we can write the stresses of these multiple layers in the objective function. We also choose the cover layer thickness (d_{Cover}), glue layer thickness (d_{Glue}), and PI substrate thickness (d_{PI}) as the optimized parameters. OLED layer and the third barrier layer (Barrier 3) are chosen as the improvement objectives; thus, the objective function F_3 is defined as following:

$$F_3 = F_1 + F_2 \quad (8)$$

$$F_1 = \left| \bar{E}_{OLED} \left[\varepsilon_{bot} + \frac{z_{OLED}}{h} (\varepsilon_{top} - \varepsilon_{bot}) \right] \right| \quad (9)$$

$$F_2 = \left| \bar{E}_{Barrier3} \left[\varepsilon_{bot} + \frac{z_{Barrier3}}{h} (\varepsilon_{top} - \varepsilon_{bot}) \right] \right| \quad (10)$$

$$\bar{E}_{OLED} = \frac{E_{OLED}}{1 - \nu_{OLED}^2} \quad (11)$$

$$\bar{E}_{Barrier3} = \frac{E_{Barrier3}}{1 - \nu_{Barrier3}^2} \quad (12)$$

Equation (8) represents the sum of the stress values on the OLED layer and the third barrier layer, as smaller as better.

4 | RESULTS AND DISCUSSION

4.1 | Simulation parameters

To verify the proposed method, the finite element simulation analysis is performed. To simulate the stress distribution of a foldable OLED panel, some mechanical parameters should be given. Table 1 presents elastic modulus and Poisson ratio of each layer in the foldable OLED panel model. Based on the products on the market, we choose the bending radius of 1.5 mm as an example for analysis.

In the simulation model, the length L of the foldable OLED panel model is 67.12 mm, the length of the rigid body L_R is 64.76 mm, and the bending radius R is 1.5 mm. The initial thicknesses of each layer are given in Table 2.

4.2 | Single-objective optimization and simulations

Based on Equation (6), the stress on OLED layer is set as the optimized objective. The optimized thickness range of cover layer, glue layer, and PI substrate are shown in Table 3.

TABLE 1 Mechanical parameters of each stacked layer in a foldable OLED panel.

Layer	E (GPa)	μ
Cover	8.6	0.29
Glue	3.2	0.45
Passivation	99.51	0.3
Cathode	70	0.33
OLED	58.6	0.3
ITO	106.5	0.3
Barrier1	99.51	0.3
Barrier2	69	0.3
Barrier3	99.51	0.3
PI	6.75	0.33
Rigid body	200	0.3

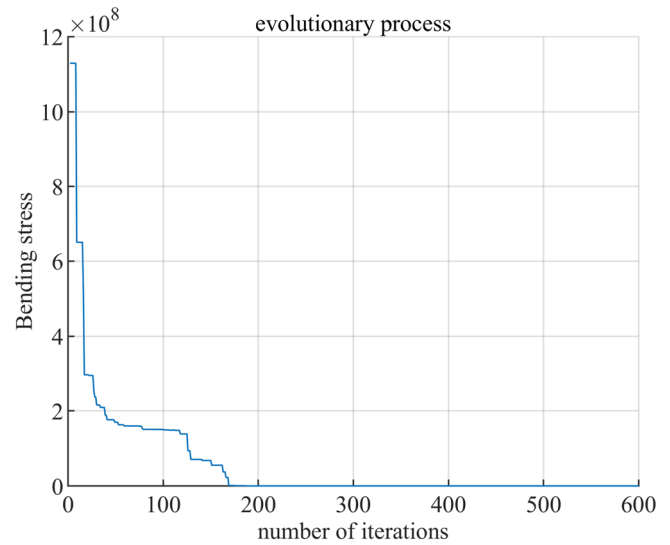
TABLE 2 The initial thickness of each layer.

Layer	Thickness (μm)
Cover	30
Glue	20
Passivation	0.4
Cathode	0.4
OLED	0.4
ITO	0.2
Barrier1	0.4
Barrier2	0.5
Barrier3	0.5
PI	18
Rigid body	20

TABLE 3 Film thickness range for single- objective optimization.

Layer	Optimal range
	Thickness (μm)
Cover	15–45
Glue	10–30
PI	9–27

The genetic algorithm employs the following operational parameters: a population size $MM = 300$, a maximum number of iterations $maxgen = 600$, a generation gap $G_{gap} = 0.9$, a crossover probability $P_c = 0.96$, and a variance probability $P_m = 0.06$. The iteration

**FIGURE 3** Relationship between objective function and number of iterations of single-objective optimization.

ends when no superior solution emerges during the evolutionary process, and the algorithm identifies the optimal solution as the smallest individual with the highest fitness value. The iteration process is depicted in Figure 3.

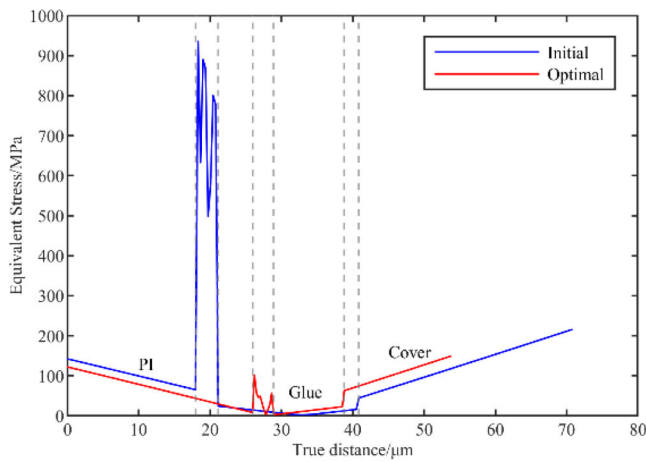
As shown in Figure 3, the objective function value decreases gradually with the increased number of genetic generations and eventually converges, which indicates that the magnitude of the objective layer stress is approaching the minimum value. After 170 generations, the objective function value tends to be constant, indicating that the stress has reached its optimal value.

The optimized parameters are calculated as follows: $d_{Cover} = 15.469 \mu\text{m}$, $d_{Glue} = 10.242 \mu\text{m}$, $d_{PI} = 26.346 \mu\text{m}$. The simulation with such configuration $d_{Cover} = 15 \mu\text{m}$, $d_{Glue} = 10 \mu\text{m}$, $d_{PI} = 26 \mu\text{m}$ is performed. Figure 4 shows the stress distributions of the foldable OLED panel along the thickness direction.

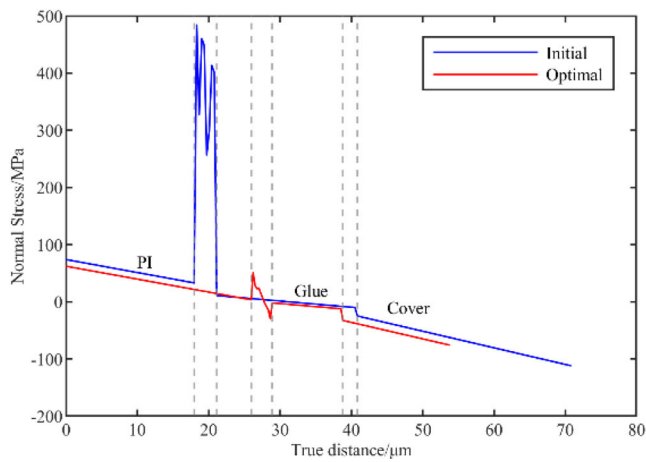
It can be seen from Figure 4 that the stresses on films are significantly reduced except for the slight increase of the stress on cover layer. The stress improvement of the films with higher elastic modulus such as passivation layer, cathode layer, OLED layer, ITO layer, and the barrier layer is particularly significant.

To better compare the simulation results of the initial and optimal thickness configuration, the stresses on passivation layer, ITO layer, cathode layer, OLED layer, and the barrier layers are taken out to present in Figure 5.

It can be seen from Figure 5 that both equivalent stresses and normal bending stresses are significant reduced with the optimal thicknesses.



(A)



(B)

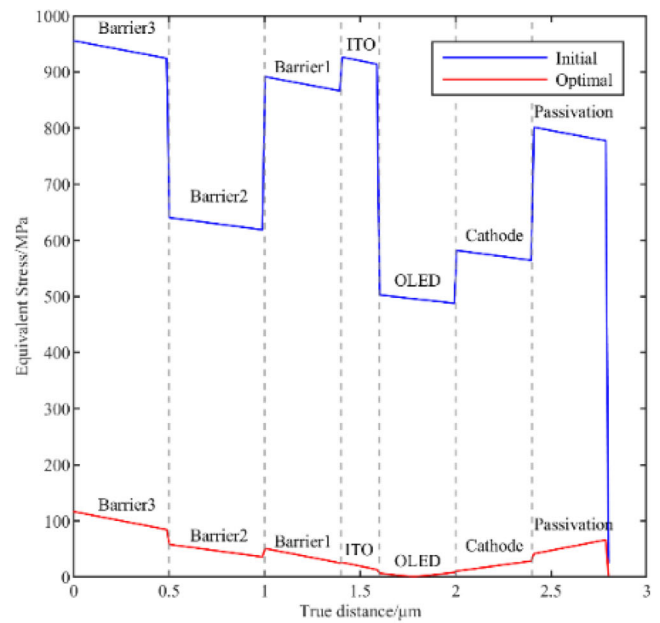
FIGURE 4 Simulation results of single-objective optimization: (A) equivalent stress distribution; (B) normal stress distribution.

4.3 | Multi-objective optimization and simulation

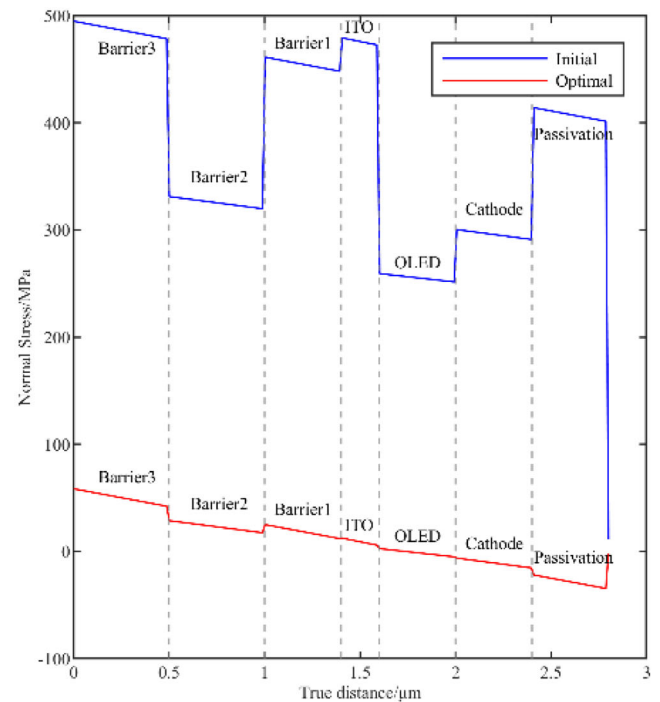
Based on Equation (8), we optimize the thicknesses of cover layer, glue layer, and PI substrate. The initial thickness of these films and the range to be optimized are the same to the single-objective optimization, as shown in Table 3.

The operational parameters are as same as single-objective optimization, except a maximum number of iterations $maxgen = 5000$. The iteration process is depicted in Figure 6.

As shown in Figure 6, the objective function value decreases gradually with the number of genetic generations and eventually converges. This trend indicates that the total stress value of the objective layers is approaching the minimum value. After 1000 generations, the



(A)



(B)

FIGURE 5 Simulation results of single-objective optimization (passivation layer, cathode layer, OLED layer, ITO layer, and barrier layer): (A) equivalent stress distribution; (B) normal stress distribution.

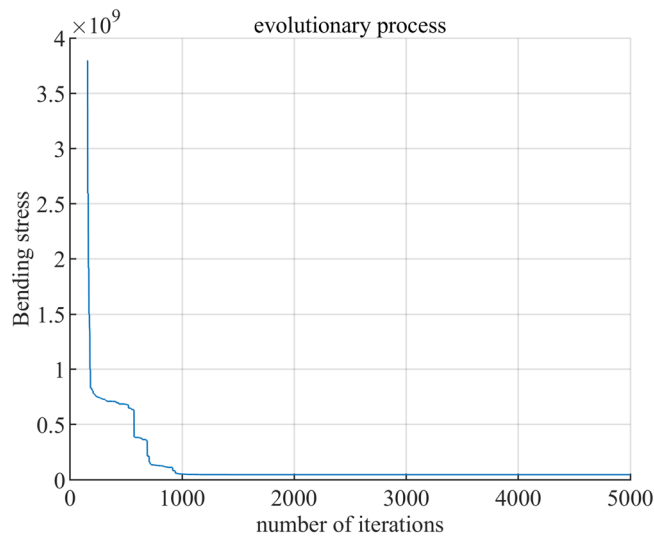
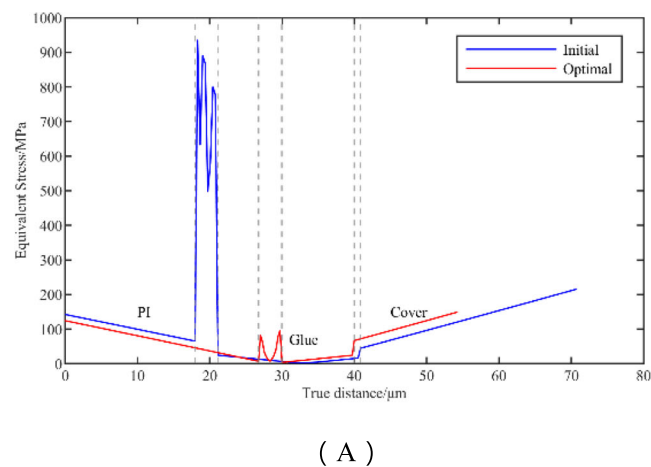
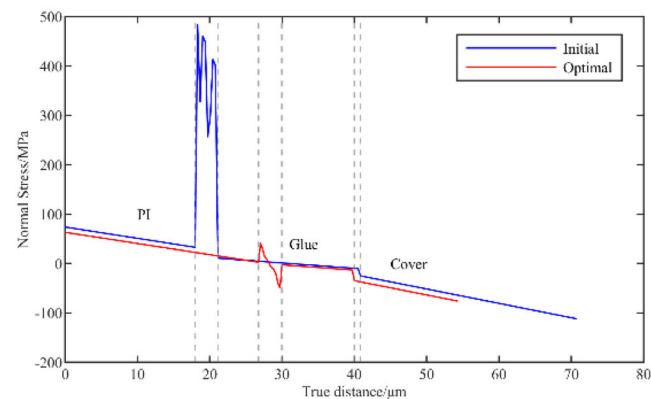


FIGURE 6 Relationship between objective function and number of iterations of multi-objective optimization.

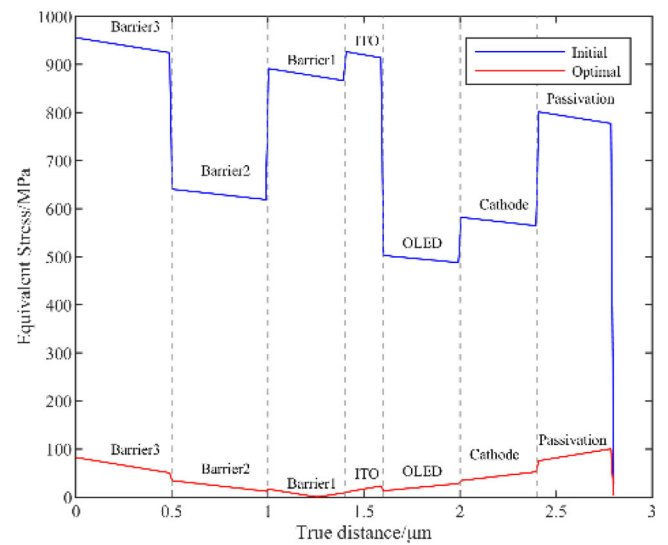


(A)

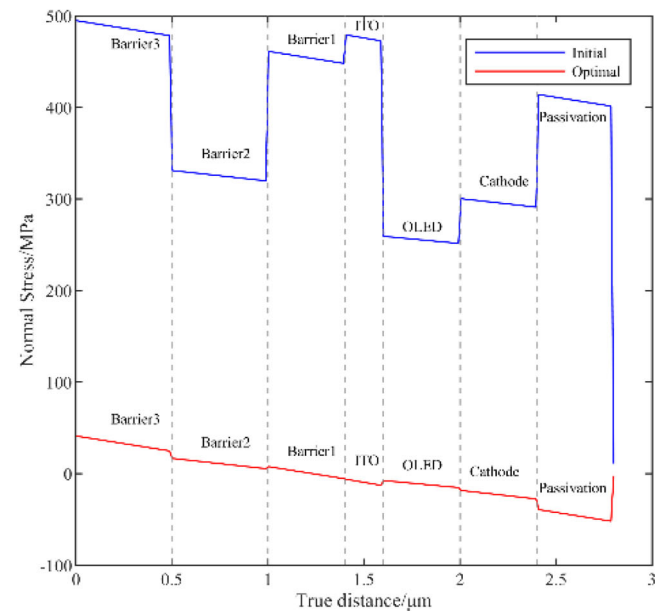


(B)

FIGURE 7 Simulation results of multi-objective optimization: (A) equivalent stress distribution; (B) normal stress distribution.



(A)



(B)

FIGURE 8 Simulation results of multi-objective optimization (passivation layer, cathode layer, OLED layer, ITO layer, and barrier layer): (A) equivalent stress distribution; (B) normal stress distribution.

objective function becomes constant, indicating that the total stress value has reached its optimal value according to our design.

The optimized parameters are calculated as follows: $d_{\text{Cover}} = 14.317 \mu\text{m}$, $d_{\text{Glue}} = 10.461 \mu\text{m}$, $d_{\text{PI}} = 26.876 \mu\text{m}$. The stress value of the objective function is calculated

according to Equation (8), and the value F_3 was obtained as 47.19 MPa. The layer thicknesses used in the simulation are as follows: $d_{Cover} = 14 \mu\text{m}$, $d_{Glue} = 10 \mu\text{m}$, $d_{PI} = 27 \mu\text{m}$. Figure 7 shows stress distributions of the foldable OLED panel along the thickness direction.

Figure 7 presents the similar trend as Figure 4. The stress distribution on passivation layer, cathode layer, OLED layer, ITO layer, and three barrier layers are also enlarged for the detailed analysis, as shown in Figure 8.

It can be seen from Figure 8 that both equivalent stresses and normal bending stresses are significant reduced. The sum of the stresses on OLED layer and the third barrier layer is 58.64 MPa. In the design aimed at reducing the stress on OLED layer independently, the sum of the stress on OLED layer and the third barrier layer is 64.82 MPa. This indicates that the stress on these two films has been further reduced.

5 | CONCLUSION

In this paper, we propose a method to optimize the structure of foldable OLED panels. Based on the calculation model for the bending stress, the genetic algorithm is employed to find the optimal configuration of each layer thickness, which can make the neutral plane happening on the key films. The stress improvement of a single layer and multiple layers are discussed, respectively. The method is verified by the numerical simulation results. The proposed method can help guide the structure design and improve folding performance of a flexible OLED panel.

ACKNOWLEDGMENTS

This research was supported by Major Science and Technology Research Project of Anhui Province under grant NO. 2023z020002.

ORCID

Qibin Feng  <https://orcid.org/0000-0002-7752-2507>

Guoqiang Lv  <https://orcid.org/0009-0000-2763-5557>

REFERENCES

- Komatsu R, Nakazato R, Sasaki T, Suzuki A, Senda N, Kawata T, et al. Repeatedly foldable book-type AMOLED display. *SID Symp Dig Tech Pap*. 2014;45(1):326–9. <https://doi.org/10.1002/j.2168-0159.2014.tb00088.x>
- Yagi I, Hirai N, Miyamoto Y, Noda M, Imaoka A, Yoneya N, et al. A flexible full-color AMOLED display driven by OTFTs. *J Soc Inf Disp*. 2008;16(1):15–20. <https://doi.org/10.1889/1.2835023>
- Niu L, Ding J, Liu W. Analysis on the mechanical behavior of flexible screens. *Materials*. 2022;15(8):2829. <https://doi.org/10.3390/ma15082829>
- Jia Y, Liu Z, Wu D, Chen J, Meng H. Mechanical simulation of foldable AMOLED panel with a module structure. *Org Electron*. 2019;65:185–92. <https://doi.org/10.1016/j.orgel.2018.11.026>
- Lee CC, Liou YY. Dependent analyses of multilayered material/geometrical characteristics on the mechanical reliability of flexible display devices. *IEEE Trans Device Mater Reliab*. 2018;18(4):639–42. <https://doi.org/10.1109/TDMR.2018.2878485>
- Walls R, Viljoen C, de Clercq H. A nonlinear, beam finite element with variable, eccentric neutral axis. *Eng Struct*. 2019; 187:341–51. <https://doi.org/10.1016/j.engstruct.2019.02.056>
- Park SK, Han JJ, Moon DG, Kim WK. Mechanical stability of externally deformed indium–tin–oxide films on polymer substrates. *Jpn J Appl Phys*. 2003;42(2R):623–9. <https://doi.org/10.1143/JJAP.42.623>
- Nematollahisarvestani A, Lee YCYC. A review on United States patents to prevent mechanical failures in foldable smartphones. *J Electron Packag*. 2021;143(2):020802. <https://doi.org/10.1115/1.4048011>
- Niu YF, Liu SF, Chiou JY, Huang CY, Chiu YW, Lai MH, et al. Improving the flexibility of AMOLED display through modulating thickness of layer stack structure. *J Soc Inf Display*. 2016;24(5):293–8. <https://doi.org/10.1002/jsid.443>
- Wang H, Hsieh M, Xie C, Cai B, Guo Y, Liu L, et al. Influence of substrate structure on the properties of flexible AMOLED displays. *SID Symp Dig Tech Pap*. 2016;47(1):1526–8. <https://doi.org/10.1002/sdtp.10990>
- Lee, CC, Wu, CS, Tzeng, TL, Tsai CH, Yeh ST, Peng YH, et al. Packaging designs and flexural stress estimation for thin-film types of OLED devices. 2012 7th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT), 2012: 415–8.

AUTHOR BIOGRAPHIES



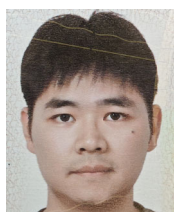
Qibin Feng received the BS and MS degrees from the Department of Automatic Instrumentation, Shanghai Machinery Institute, Shanghai, China, in 1992 and 1995, respectively, and the Ph.D. degree from the Department of Information Engineering, Vienna University of Technology, in 2006. She was a teacher with Hefei University of Technology from 1995 to 2001. She became a senior researcher and an associate professor of Academy of Photoelectric Technology, Hefei University of Technology. Her research interests include LC display, LED backlight, 3D display.



Chen Zheng is currently pursuing the MEng degree at Hefei University of Technology. His current research interests include mechanical simulation of flexible OLED displays.



Zhe Dong is currently pursuing the MEng degree at Hefei University of Technology. Her current research interests include mechanical simulation of flexible TFTs.



Yusong Guo is currently pursuing the MEng degree at Hefei University of Technology. His current research interests include volume holographic grating and diffractive waveguide.



Zi Wang received the BS and Ph.D. degrees from the Department of Optics and Optical Engineering, University of Science and Technology of China, Hefei, China, in 2012 and 2017, respectively. Since then, he

worked in the Academy of Photoelectric Technology, Hefei University of Technology as an assistant researcher. His current research interests include LC display, LED backlight, and 3D display.



Guoqiang Lv received the BS and MS degrees from the Department of Optical Instrumentation, Zhejiang University, Hangzhou, China, in 1983 and 1986, respectively. Since then, he worked in the school of instrumentation, Hefei University of Technology, and became a professor in 2001. He was a Senior Visiting Scholar with the Institute of Computer Technology, Vienna University of Technology, from 2001 to 2002. Since 2003, he has been the dean of Academy of Photoelectric Technology, Hefei University of Technology. His current research interests include LC display, LED backlight, and 3D display.

How to cite this article: Feng Q, Zheng C, Dong Z, Guo Y, Wang Z, Lv G. Genetic algorithm-based structural optimization of foldable OLED panels. *J Soc Inf Display*. 2025;33(4):255–62. <https://doi.org/10.1002/jsid.2032>